

Q1.c

```
#include <stdio.h>
#include <math.h>
#include <time.h>

void f1(int n){
    if(n<=1) return;

    int sz=n*log(n);
    double c=0.00;
    for(int i=0;i<sz;i++){c++;}
    f1(n/2);
    f1(n/2);
}

int main(){
    int ar[]={500,1000,2000,4000,8000};
    int size=5;
    for(int i=0;i<size;i++){
        int num=ar[i];
        clock_t st=clock();
        f1(num);
        clock_t sp=clock();
        double t =(double) (sp-st)/CLOCKS_PER_SEC;
        printf("%-15d | %lf \n",num,t);
    }

    return 0;
}
```

[q1.py](#)

```
import matplotlib.pyplot as plt

# Data
x = [500, 1000, 2000, 4000, 8000]
y = [0.000167, 0.000353, 0.000853, 0.002037, 0.004894]

# Plot
plt.plot(x, y, marker='o')
plt.xlabel("Input Size")
plt.ylabel("Time (seconds)")
plt.title("Input Size vs Time")
plt.grid(True)
plt.show()
```

Q2.c

```
#include <stdio.h>

#include <stdlib.h>

#include <time.h>

void linear(int *array, int size, int target) {

    clock_t st = clock();

    for (int i = 0; i < size; i++) {

        if (array[i] == target) {

            break;

        }

    }

}
```

```
clock_t stop = clock();

double time_taken = (double)(stop - st) / CLOCKS_PER_SEC;

printf("%-15d | %lf for linear\n", size, time_taken);
}

void binary(int *array, int size, int target, int left, int right) {

    clock_t st = clock();

    while (left <= right) {

        int mid = left + (right - left) / 2;

        if (array[mid] == target)

            break;

        else if (target > array[mid])

            left = mid + 1;

        else

            right = mid - 1;

    }

    clock_t sp = clock();

    double time_taken = (double)(sp - st) / CLOCKS_PER_SEC;
```

```

    printf("%-15d | %lf  for binary\n", size, time_taken);
}

void bubble(int *array, int size) {
    clock_t st = clock();

    for (int i = 0; i < size - 1; i++) {
        for (int j = 0; j < size - i - 1; j++) {
            if (array[j] > array[j + 1]) {
                int temp = array[j];
                array[j] = array[j + 1];
                array[j + 1] = temp;
            }
        }
    }

    clock_t sp = clock();

    double time_taken = (double)(sp - st) / CLOCKS_PER_SEC;

    printf("%-15d | %lf  for bubble\n", size, time_taken);
}

int main() {
    int sizes[] = {500, 1000, 2000, 4000, 8000};

```

```
int n = sizeof(sizes) / sizeof(sizes[0]);

int target = -1;

for (int i = 0; i < n; i++) {

    int *ar = malloc(sizes[i] * sizeof(int));

    for (int j = 0; j < sizes[i]; j++)

        ar[j] = j;

    linear(ar, sizes[i], target);

    bubble(ar, sizes[i]);

    binary(ar, sizes[i], target, 0, sizes[i] - 1);

    printf("\n");

    free(ar);

}

return 0;

}
```

[q2.py](#)

```
import matplotlib.pyplot as plt

# Input sizes
x = [500, 1000, 2000, 4000, 8000]

# Time data (seconds)
linear = [0.000002, 0.000002, 0.000008, 0.000003, 0.000011]
bubble = [0.000270, 0.001595, 0.006781, 0.021474, 0.029585]
binary = [0.000001, 0.000002, 0.000000, 0.000000, 0.000000]

# Plot
plt.plot(x, linear, marker='o', label="Linear Search")
plt.plot(x, bubble, marker='o', label="Bubble Sort")
plt.plot(x, binary, marker='o', label="Binary Search")

plt.xlabel("Input Size")
plt.ylabel("Time (seconds)")
plt.title("Algorithm Time Complexity Comparison")
plt.legend()
plt.grid(True)
plt.show()
```

